

Signifyd INTEGRATION

Version 3.5.0



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1. Intended Audience

This document is intended for the technical audience that will be directly involved in the setup and/or integration of this Salesforce Commerce Cloud cartridge.

2. Summary

Signifyd is a fraud solution that provides a financial guarantee, allowing businesses to increase sales while reducing fraud losses. The Signifyd cartridge will be integrated into Salesforce Commerce Cloud using three primary API integration points:

1. Pre-payment authorization (Checkout API, Transaction API)
2. Post-payment authorization (Sale API, Webhooks)
3. Post order fulfilment (Fulfilment API)

Sale API is used if merchants are executing a "Post-Auth" Flow where they call into Signifyd after the order has been placed and the payment has been authorized with the Payment Gateway. Checkout API is used if merchants are executing a "Pre-Auth" Flow where they call into Signifyd before authorizing a payment with the Payment Gateway. In "Pre-Auth" flow, for each order, there are at least 2 API calls, a Checkout API call prior to payment authorization and a Transaction API after authorization

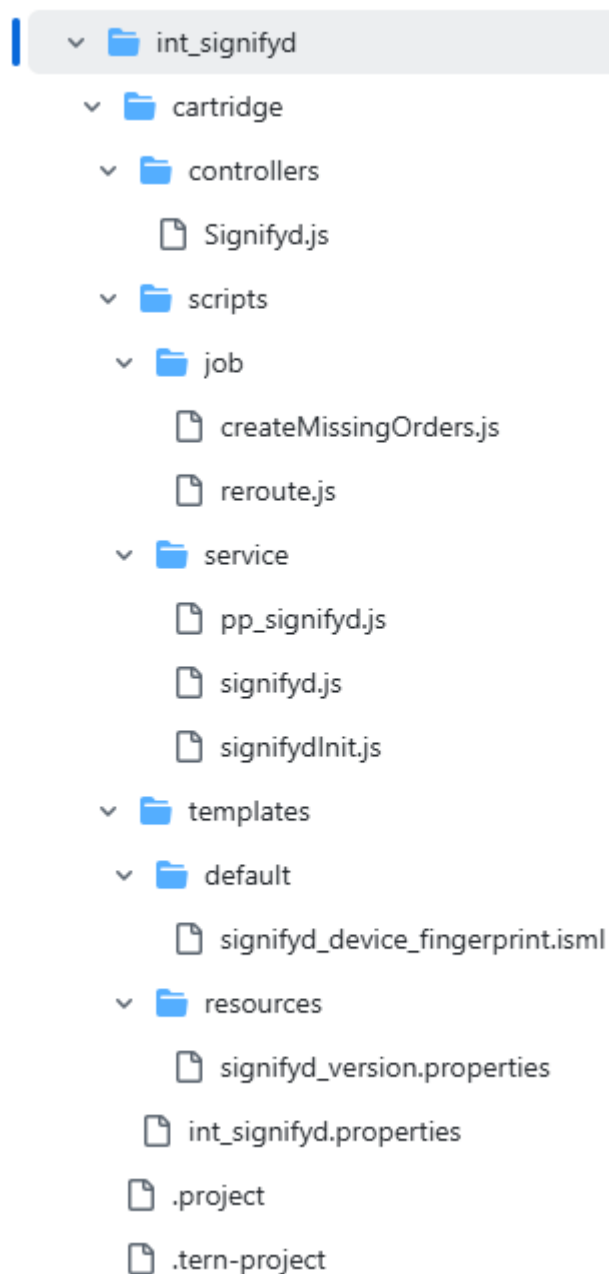
This document primarily serves as the LINK implementation guide for setting up Signifyd on SFRA and SiteGenesis.

The setup and custom code configuration described in this document assumes the use of SFRA version $\geq 3.3.0$ and SiteGenesis 103.1.11 release of app_storefront_core. Custom coding might be required if adapting the cartridge to work with other SiteGenesis releases, pre-2.0 releases, and versions of SiteGenesis that do not include the RequireJS framework.

3. Components

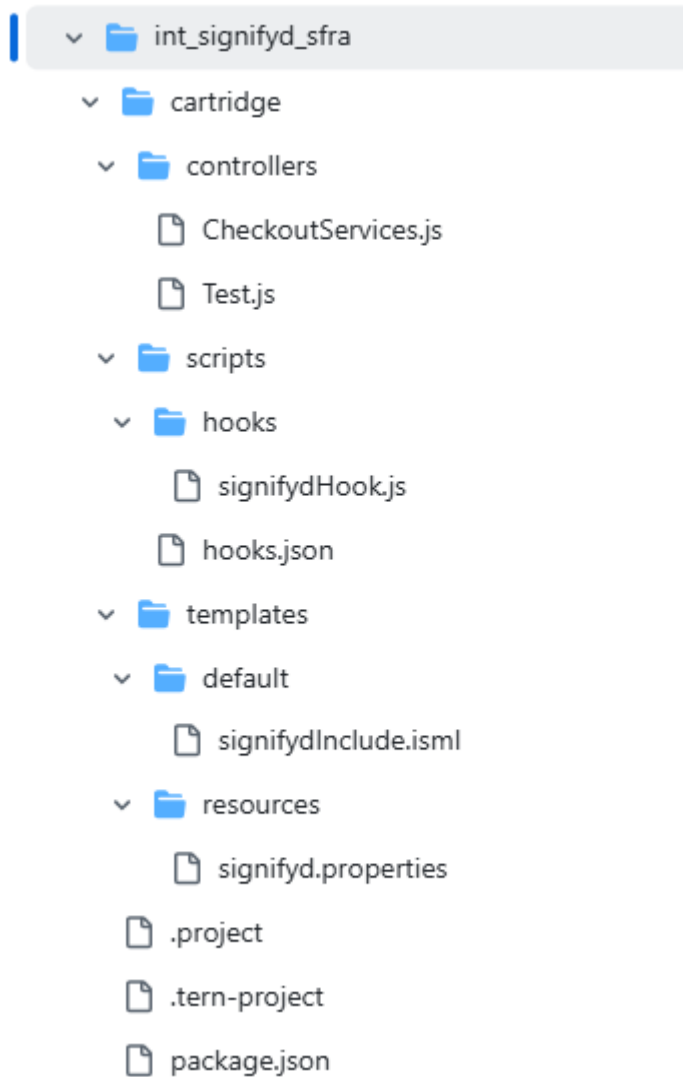
3.1 int_signifyd

The **int_signifyd** cartridge contains the base changes for the cartridge that are common for both SFRA and SiteGenesis.



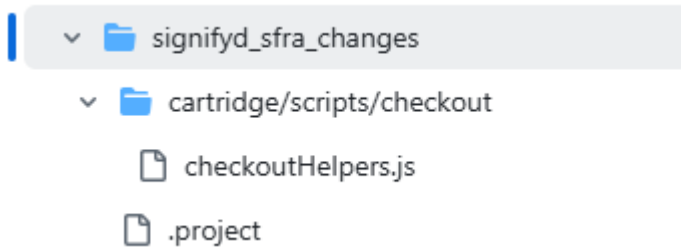
3.2 int_signifyd_sfra

The **int_signifyd_sfra** cartridge contains files specific to the SFRA integration.



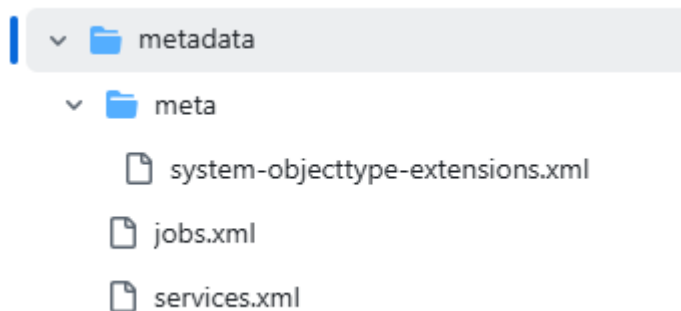
3.3 signifyd_sfra_changes

The **signifyd_sfra_changes** cartridge contains examples of files that should be changed on the SFRA base file or customized file used by the merchant integration.



3.4 Metadata

The cartridge metadata includes System Objects, such as Order and Site Preferences, along with the necessary jobs and services essential for the cartridge's functionality.



3.4.1 System Object Types

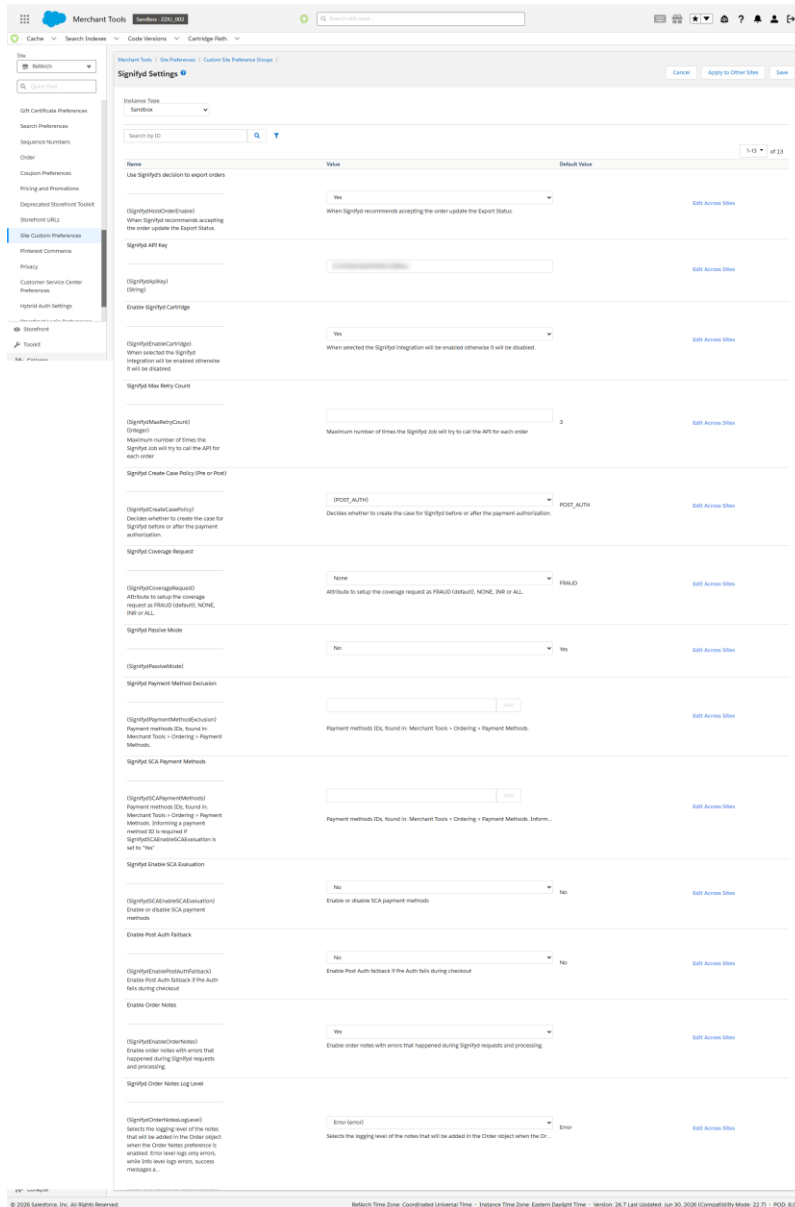
3.4.1.2 Order

The Order system object is updated with Signifyd related custom attributes, visible in the Signifyd custom attribute group.

ID	Name	Type	Attribute Settings
SignifydCaseID	Signifyd Case ID	String	
SignifydFraudScore	Signifyd Fraud Score	Integer	
SignifydOrderURL	Signifyd Order URL	String	
SignifydRetryCount	Signifyd Retry Count	Integer	
SignifydPolicy	Signifyd Policy	String	
SignifydPolicyName	Signifyd Policy Name	String	
SignifydOrderFailedReason	Signifyd Order Failed Reason	String	
SignifydPaymentMethodExclusionFlag	Signifyd Payment Method Exclusion Flag	Boolean	
SignifydSCAOutcome	Signifyd SCA Outcome	String	

3.4.1.3 Site Preferences

The Signifyd Custom Site Preferences are inside the group Signifyd Settings. The preferences are used to control the cartridge behavior.



4. Functional Overview

4.1 Features

4.1.1 Post-authorization

If the custom site preference **SignifydCreateCasePolicy** is set to “POST_AUTH”, Signifyd decisions are returned asynchronously, so an HTTP callback (webhook) is used to return their guarantee decision. Refer to section 7 - Process Flow Diagrams for post-auth process flow.

(SignifydCreateCasePolicy)

Decides whether to create the case for Signifyd before or after the payment authorization.

(POST_AUTH)

Decides whether to create the case for Signifyd before or after the payment authorization.

POST_AUTH

Action 1. The Signifyd Sale API is called after the Salesforce Commerce Cloud place order event, which happens after the order has gone through the authorization process against the payment provider right before displaying the order summary page. Because it is only called during the order creation, this will ensure that Create Case is never called again for that same order. The Sale API is called when all below actions are completed:

1. Payment is authorized without errors.
2. The order is successfully placed.
3. Confirmation of the order is sent to the customer.

Note: Any customizations done by the merchant should guarantee that the Sale API is called after the above steps are completed.

Action 2. The second integration point is a publicly accessible URL that will be used as the callback/web-hook endpoint. This endpoint will be called when Signifyd has completed its fraud assessment, and a decision is made to either approve or decline the order for financial guarantee. This triggers an update to the order in SFCC and could also indicate that the order is ready to export (depending on settings). In case of failure during the callback function execution that prevents the SFCC order object to be updated (for example an 'optimistic locking failure'), the cartridge will retry 3 times.

Note: In the case that Signifyd returns a decline decision, the merchant should decide how to proceed with that order. If the order is manually cancelled in SFCC and/or with the payment processor, communication to the client should also be done by the merchant, since the checkout experience on the storefront itself is complete at this stage.

4.1.2 Pre-authorization

If the custom site preference **SignifydCreateCasePolicy** is set to "PRE_AUTH", Signifyd decisions are returned synchronously after the relevant Signifyd API is called. Refer to section 7 - Process Flow Diagrams for pre-auth process flow.

(SignifydCreateCasePolicy)

Decides whether to create the case for Signifyd before or after the payment authorization.

(PRE_AUTH)

Decides whether to create the case for Signifyd before or after the payment authorization.

POST_AUTH

Action 1. The Signifyd Checkout API is called before the Salesforce Commerce Cloud order has gone through the authorization process against the payment provider. If Signifyd returns a declined/rejected response, the authorization call will not happen, and the order will be failed. The failed order will be in "failed" status in Business Manager, where the merchant can also check the Signifyd order attributes, like the score received and the reason it failed. The storefront will display a default error message to the customer. If the merchant wants to customize this message, it can be changed directly in the code. If Signifyd returns an accept/approved response, Salesforce Commerce Cloud proceeds with the authorization process against the payment provider.

Action 2. For Signifyd "Approved" orders, the Transaction API is called to send payment status after the authorization result. Some of the fields should be manually added by the merchant for the Transaction API. More details in the section 3.1.11 API Integration – Limitations and Constraints.

4.1.3 Passive Mode

Passive mode can be used by setting the custom site preference **SignifydPassiveMode** to "Yes". If passive mode is enabled, Signifyd decision will be visible on Business Manager through the order custom attributes, but it won't impact the order status regardless of Signifyd accept/reject decisions. When the passive mode is switched off, Signifyd decision will be updated on the order, and the order status will be impacted according to the accept/reject decisions.

4.1.4 Post-Auth Fallback

When in Pre-Auth mode, the merchant can set the **SignifydEnablePostAuthFallback** preference to "Yes" to have a fallback Post-Auth call in case the original Pre-Auth call fails during checkout. By default, this preference is set to "No".

Enable Post Auth Fallback

(SignifydEnablePostAuthFallback)
Enable Post Auth fallback if Pre Auth fails during checkout

Yes

No

Enable Post Auth fallback if Pre Auth fails during checkout

4.1.5 Order Fulfillment

Fulfillment represents a shipment of one or more items in an order. The merchant can submit fulfillment details for orders that were shipped, even if the order is partially shipped. The function **sendFulfillment** from the file **signifyd.js** is available to be called at the time when the order is fulfilled on the merchant order flow. Some of the fields for the Fulfillment API need to be added manually by the merchant. More details in the section 5.5 Limitations and Constraints.

```
function getSendFulfillmentParams(order) {
    var fulfillmentStatus = order.getShippingStatus().displayValue === "PARTSHIPPED" ? "PARTIAL" : "COMPLETE";
    var shipments = order.getShipments();

    var paramsObj = {
        orderId: order.orderNo,
        fulfillmentStatus: fulfillmentStatus,
        fulfillments: []
    };

    var iterator = shipments.iterator();
    while (iterator.hasNext()) {
        var shipment = iterator.next();
        paramsObj.fulfillments.push({
            shipmentId: shipment.shipmentNo,
            // shipmentStatus: "", // to be updated by the merchant
            // trackingId: "", // to be updated by the merchant
            // trackingNumbers: "", // to be updated by the merchant
            // carrier: "", // to be updated by the merchant
            products: getProductLineItems(shipment.productLineItems),
            destination: {
                fullName: shipment.shippingAddress.fullName,
                organization: shipment.shippingAddress.companyName,
                address: getDeliveryAddress(shipment),
                confirmationPhone: shipment.shippingAddress.phone
            }
        });
    }

    return paramsObj;
}
```

4.1.6 Reroute

The **sendReroute** function is available to be called whenever the delivery address on an order needs to be changed. The change will be linked to the original order with the `orderId` parameter in the body. The **getSendRerouteParams** auxiliary function creates the request with the necessary information for the API call.

```
function getSendRerouteParams(order) {
    var orderShipments = order.getShipments();
    var shipments = [];

    if (!empty(orderShipments)) {
        var iterator = orderShipments.iterator();
        while (iterator.hasNext()) {
            var shipment = iterator.next();
            shipments.push({
                shipmentId: shipment.shipmentNo,
                destination: {
                    fullName: shipment.shippingAddress.fullName,
                    organization: shipment.shippingAddress.companyName,
                    address: {
                        streetAddress: shipment.shippingAddress.address1,
                        unit: shipment.shippingAddress.address2,
                        city: shipment.shippingAddress.city,
                        provinceCode: shipment.shippingAddress.stateCode,
                        postalCode: shipment.shippingAddress.postalCode,
                        countryCode: shipment.shippingAddress.countryCode.value
                    }
                }
            });
        }
    }

    var paramsObj = {
        orderId: order.orderNo,
        shipments: shipments
    }

    return paramsObj;
}

function sendReroute(orderId) {
    var order = OrderMgr.getOrder(orderId);

    if (EnableCartridge) {
        if (order && order.currentOrderNo) {
            try {
                var params = getSendRerouteParams(order);
                var service = signifydInit.sendReroute();
            }
        }
    }
}
```

4.1.6 Custom Preferences

The Signifyd cartridge has a configuration setting to hold the order or immediately export depending on Signifyd's guarantee decision. The site preference setting is called **SignifydHoldOrderEnable** and if set to true this indicates that the Salesforce Commerce Cloud order is held until the webhook listener is called and indicates that the order is approved. This is accomplished by marking the order as 'Not Exported' to prevent it from exporting until the webhook listener updates the order to 'Ready for Export'.

The site preference **SignifydCreateCasePolicy** decides whether the create case policy will be pre-authorization (synchronous) or post-authorization (asynchronous).

Use the site preference **SignifydCoverageRequest** to setup the Coverage Request as FRAUD (default), NONE, INR or ALL.

4.1.7 Manual Review

To use Signifyd's manual review feature for pre-auth, you would need to configure the `ORDER_CHECKPOINT_ACTION_UPDATE` webhook. In this scenario, Signifyd decisions (for manual review orders) will be returned asynchronously, so an HTTP callback (webhook) is used to return the guarantee decision.

The expected order flow process for manual review orders in pre-auth setting is mentioned below:

1. Order is sent to Signifyd for pre-auth fraud check
2. Signifyd deems the order to be a manual review order and returns a PENDING decision response back to SFCC (instead of ACCEPT or REJECT)
3. SFCC sends the order for payment authorization
4. Once the payment authorization process is complete, SFCC sends the payment authorization details to Signifyd via the transaction API endpoint
5. Based on the payment authorization details sent, Signifyd makes a manual decision on the order (order moved from Pending to Accept/Reject)
6. ORDER_CHECKPOINT_ACTION_UPDATE webhook is sent to SFCC for this asynchronous decision by Signifyd
7. SFCC consumes Signifyd's ORDER_CHECKPOINT_ACTION_UPDATE webhook and processes the order as per the defined merchant settings. Example:
 - a. For Signifyd approved orders, the order may be exported to merchant's OMS
 - b. For Signifyd declined/rejected orders, the order may not be exported to merchant's OMS

The table below lists out the different types of scenarios related to Manual Review orders that a merchant may want to consider.

Scenario	Signifyd decision for a Manual Review Order	Merchant Scenario	Action needed from Merchant	Who takes liability?
1	Accept	Merchant decides to progress order to fulfillment	No action needed as this should happen automatically via ORDER_CHECKPOINT_ACTION_UPDATE webhooks	Signifyd
2	Reject	Merchant decides to fulfill Signifyd declined order	i. Merchant user would need to mark the order as Good within Signifyd console ii. Manually progress order to OMS for order fulfillment	Merchant
3	Reject	Merchant decides to not fulfill Signifyd declined order	- Merchant would need to reverse payment authorization for this order since the authorization was already done. - In case the payment was captured prior to receiving Signifyd's decision, the payment would need to be refunded	N/A
4	Reject	Merchant decides to resubmit order to Signifyd	- Merchant users would need to resubmit order from the console and add additional details related to why this declined order is being resubmitted. - If Signifyd subsequently approves this order post resubmit, then order should be progressed to the OMS for fulfillment (this would happen automatically via webhooks) - If Signifyd declines this order again post resubmit, then merchant can	

			decide if they still want to progress the order (scenario 2) or not (scenario 3)	
--	--	--	--	--

4.1.8 Webhooks

Webhooks are sent by Signifyd when Signifyd approves/declines an order for guarantee and will be indicated in the checkpointAction string as "ACCEPT", "REJECT", or "HOLD". Typically, for pre-auth integrations, there is no need to configure a webhook as Signifyd's decision will be synchronous.

However, if a merchant wants to use Signifyd's manual review feature for pre-auth, then you would need to configure the ORDER_CHECKPOINT_ACTION_UPDATE webhook. In this scenario, Signifyd decisions (for manual review orders) will be returned asynchronously, so an HTTP callback (webhook) is used to return the guarantee decision.

Please note: For all Signifyd APIs, "caseId" = "signifydId", which is Signifyd's unique identifier for the order in our systems.

Instructions to setup V3 Webhooks are available on this page:

<https://docs.signifyd.com/#operation/CreateTeamWebhook>

Once the Signifyd decision has been made, it will be listed under order attributes.

4.1.9 Payment Exclusion

Merchants can add payment methods to an exclusion list by adding the payment method ids in the custom preference **SignifydPaymentMethodExclusion**. The payment method id informed in the preference should be the same id visible in the Business Manager menu **Ordering > Payment Methods**.

Signifyd Payment Method Exclusion

(SignifydPaymentMethodExclusion)

Payment methods IDs, found in:
Merchant Tools > Ordering > Payment
Methods.

PAY PAI 

Payment methods IDs, found in: Merchant Tools > Ordering > Payment Methods.

Orders with an excluded payment method from the preference will be ignored during checkout and will not trigger a request to Signifyd. These orders will instead be directly sent for payment authorization. The boolean attribute **SignifydPaymentMethodExclusionFlag** in the order will indicate that this order was ignored by Signifyd.

4.1.10 Create Missing Orders Job

The **Signifyd-CreateMissingOrders** job can be scheduled by the merchant to run periodically and catch orders that do not have a case ID (SignifydCaseID attribute = null), have a retry count less than the maximum retry count configured in the custom preference **SignifydMaxRetryCount** (default is 3), and do not have the status FAILED or CANCELLED. Additionally, orders that were ignored due to the payment method being listed in the payment exclusion custom preference and have the attribute **SignifydPaymentMethodExclusionFlag** set as true, will not be processed by the job.

4.1.11 Order notes

Merchants can enable the custom site preference **SignifydEnableOrderNotes** to add order notes related to Signifyd errors and information messages, in addition to the normal logging. When this functionality is enabled, the preference **SignifydOrderNotesLogLevel** can be used to control if the order notes will be added only for errors or for general information.

Enable Order Notes (SignifydEnableOrderNotes) Enable order notes with errors that happened during Signifyd requests and processing.	Yes Enable order notes with errors that happened during Signifyd requests and processing.
Signifyd Order Notes Log Level (SignifydOrderNotesLogLevel) Selects the logging level of the notes that will be added in the Order object when the Order Notes preference is enabled. Error level logs only errors, while Info level logs errors, success messages and other general information.	Error (error) Selects the logging level of the notes that will be added in the Order object when the Order Notes preference is enabled. Error level logs only errors, while Info level logs errors, success messages and other general information.

5. Integration

5.1 Metadata Import

5.1.1 System Object Type Extensions

First step is to import system object definitions for the Signifyd attributes for Order and Site Preferences. These are provided with the cartridge in the **metadata/meta/system-objecttype-extensions.xml** file.

Upload this file via Business Manager into your site:

1. Click on the Upload" button in Administration > Site Development > Import & Export Site Development

Site Development > Import & Export

Import & Export

Meta Data

[Import](#) and [export](#) your system meta data (i.e., system type extensions, custom object types, custom preference definitions).

Import

Export

Geolocations

[Import](#) geolocations for a country.

Import

Import & Export Files

[Upload](#) and [download](#) your import and export files.

Upload

Download

2. Choose your local file and click on "Upload"

Upload Import Files

Upload File:

Choose File system-obje...tensions.xml

Upload

3. Click on the Back button to return to the Import page.

4. On the Meta Data section click on the Import button

Import & Export

Meta Data

[Import](#) and [export](#) your system meta data (i.e., system type extensions, custom object types, custom preference definitions). [Import](#) [Export](#)

Geolocations

[Import](#) geolocations for a country. [Import](#)

Import & Export Files

[Upload](#) and [download](#) your import and export files. [Upload](#) [Download](#)

5. Select the **system-objecttype-extensions.xml** file that you just uploaded and click 'next' to go through the import process.

System Type Extension Import - Select File

Step 1 of 3. Next Step: Validate Import File

The list below shows all uploaded import files. Please select the file that you want to import.

Select	Name	File Size	Last Modified
☒	system-objecttype-extensions.xml	13.53 KB	11/29/23 12:07:32 pm

[Next >>](#) [Cancel](#)

Showing 1 - 1 of 1 items

You will now see a 'Signifyd Settings' attribute group in the site preference section. Site Preferences > Site Custom Preferences:

Custom Site Preference Groups ⓘ

 Search by ID

Find



ID	Name	Description	Preferences
Storefront Configs	Storefront Configurations		14
Signifyd Settings	Signifyd Settings		11

You will now also be able to see the Signifyd attribute under an order by clicking on the 'Attributes' tab.

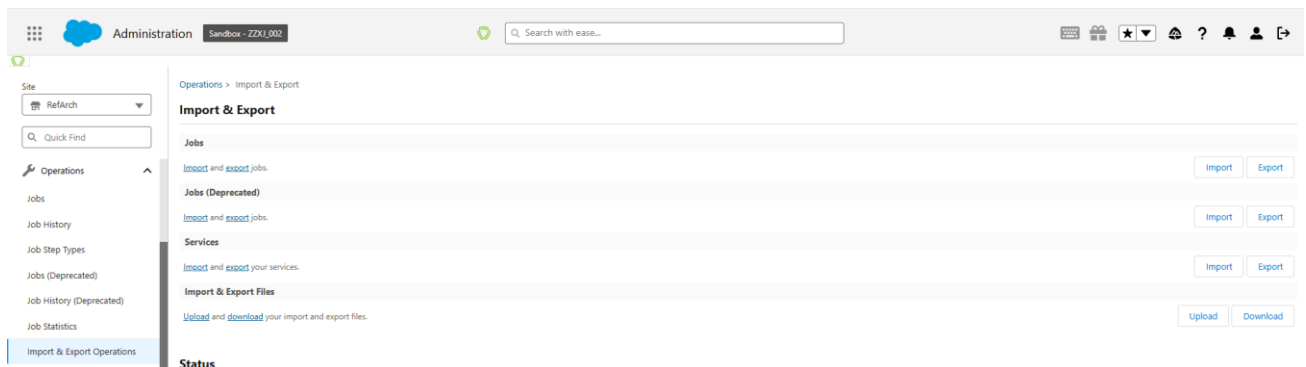
Attributes for Order '00015203'

On this page you can edit the attributes of the order. Fields with a red asterisk (*) are mandatory. Click **Apply** to save changes. Click **Reset** to revert your changes.

Signifyd	
 Signifyd Case ID:	7043955649
 Signifyd Fraud Score:	979
 Signifyd Order URL:	https://www.signifyd.com/cases/7043955649
 Signifyd Retry Count:	1
 Signifyd Policy:	accept
 Signifyd Policy Name:	SIGNIFYD_APPROVED
 Signifyd Order Failed Reason:	No data is available
 Signifyd Payment Method Exclusion Flag:	☐
 Signifyd SCA Outcome:	No data is available
 Signifyd Request:	No data is available
 Signifyd Response:	No data is available
 Signifyd Placement:	No data is available

5.1.2 Services

In Business Manager go to Administration->Operations->Import Export and select 'upload' then browse to the **services.xml** file located in the meta folder included with the Signifyd cartridge.

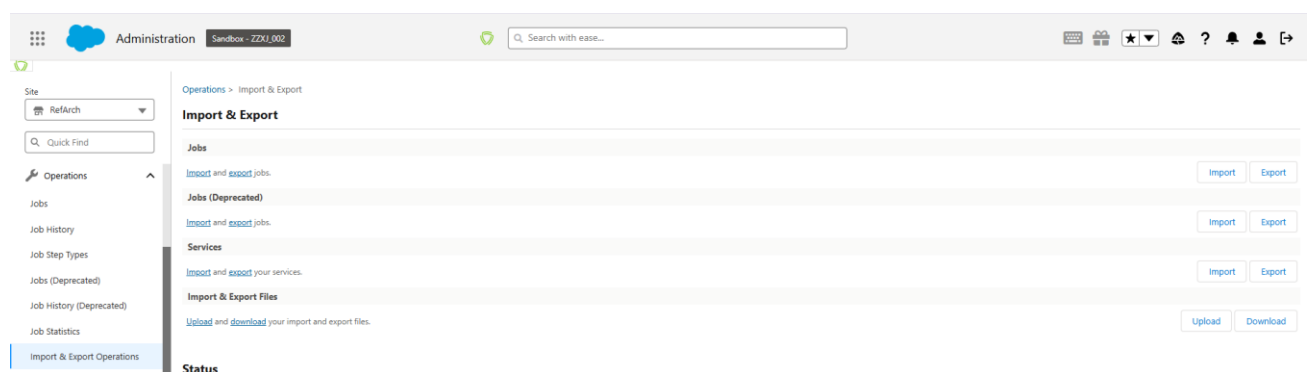


Once the services file is uploaded click Import and choose **Merge** to import the default Service Framework configuration. Navigate to the Signifyd services configuration and make sure they are enabled.

☐	SignifydCheckout	HTTP	SignifydCheckout	SignifydCheckout	Live
☐	SignifydReroute	HTTP	Signifyd Profile	SignifydReroute	Live
☐	SignifydSale	HTTP	Signifyd Profile	SignifydSale	Live
☐	SignifydSendFulfillment	HTTPForm	Signifyd Profile	signifyd.rest.send.fulfillment.development.cred	Live
☐	SignifydTransaction	HTTP	Signifyd Profile	SignifydTransaction	Live

5.1.3 Jobs

In Business Manager, go to Administration -> Operations -> Import Export and under Jobs select 'upload' then browse to the *jobs.xml* file located in the meta folder included with the Signifyd cartridge.



Once the jobs file is uploaded, click on Import and choose **Replace** to import the default Job Schedules configuration.

In Business Manager go to Administration->Operations->Job Schedules. The Signifyd-CreateMissingOrders job will be displayed:

☐	Signifyd-CreateMissingOrders	-	RefArch	-	✓	
☐	Signifyd-Reroute	-	RefArch	-	⌚	

Select Signifyd-CreateMissingOrders to enter the Job Schedule configuration. Configure your Job Schedule to run once, daily, or on any desired schedule. We recommend that you schedule your job to run at least once a day.

Signifyd-CreateMissingOrders ?

General Schedule and History Resources Job Steps Failure Handling Notification

☒ Enabled

Active

Trigger

Recurring Interval

5.2 SFRA Integration

5.2.1 Script changes

Script: checkoutHelpers.js

Path: signifyd_sfra_changes/cartridge/scripts/checkout/checkoutHelpers.js

Function **placeOrder()** on **checkoutHelpers.js** was modified to add the following logic:

If the cartridge is enabled and the **SignifydHoldOrderEnable** is set to **Yes**, then the order export status will be set to Not Exported and will later be updated based on Signifyd's webhook decision.

If the cartridge is enabled and the **SignifydHoldOrderEnable** is set to **No**, then the order export status will be set to Ready For Export as in the default **placeOrder()** function.

Different logic can be added if the merchant wants to have customization for specific payments.

5.2.2 Adding the necessary changes

Case 1 - Merchant didn't customize the base SFRA checkoutHelpers.js file

The file **checkoutHelpers.js** contains code to override the SFRA default function **placeOrder()** and add custom logic to set the export status according to the **SignifydHoldOrderEnable** custom preference.

The file should be placed on the following path, to extend the existing SFRA file:

{merchant_customized_cartridge}/cartridge/scripts/checkout/checkoutHelpers.js

Case 2 - Merchant already customized the base SFRA checkoutHelpers.js file

If the merchant already has custom logic added on his own checkoutHelpers.js file, only the necessary changes can be added (example file available on signifyd_sfra_changes/cartridge/scripts/checkout/checkoutHelpers.js):

```

/* Signifyd Modification Start */
var signifyEnabled = dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydEnableCartridge');
var signifydHoldOrderEnable = dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydHoldOrderEnable');

if (signifyEnabled) {
    ....if (signifydHoldOrderEnable === true) {
    ....|...order.setExportStatus(Order.EXPORT_STATUS_NOTEXPORTED);
    ....|...} else {
    ....|...order.setExportStatus(Order.EXPORT_STATUS_READY);
    ....|...}
    ....} else {
    ....|...order.setExportStatus(Order.EXPORT_STATUS_READY);
    ....|...}
}
/* Signifyd Modification End */

```

5.3 SiteGenesis Integration – Controllers

5.3.1 Modifications in System Controller

Enabling SFCC to send requests to Signifyd requires a modification to the system controller file:
controllers/COSummary.js

Add these two rows in the beginning of the **submit** method:

```

var Signifyd = require('int_signifyd/cartridge/scripts/service/signifyd');
var orderSessionID = Signifyd.getOrderSessionId();

```

Add those two rows near the end of the same function method:

```

Signifyd.setOrderSessionId(placeOrderResult.Order, orderSessionID);
Signifyd.Call(placeOrderResult.Order);

```

Those two last lines (number 2) should go within the last “else if” statement.

```

else if (placeOrderResult.order_created) {
....<Insert code here>.....
}

```

The final code should look like this:

```

function submit() {
    ...// Calls the COPlaceOrder controller that does the place order action and
    ...// COPlaceOrder returns a JSON object with an order_created key and a bool
    ...// If the order creation failed, it returns a JSON object with an error key

    .../* Signifyd Modification Start */
    ...var Signifyd = require('int_signifyd/cartridge/scripts/service/signifyd');
    ...var orderSessionID = Signifyd.getOrderSessionId();
    .../* Signifyd Modification End */
    ...
    ...var placeOrderResult = app.getController('COPlaceOrder').Start();
    ...if (placeOrderResult.error) {
    ...    start({
    ...        PlaceOrderError: placeOrderResult.PlaceOrderError
    ...    });
    ...} else if (placeOrderResult.order_created) {
    ...    /* Signifyd Modification Start */
    ...    Signifyd.setOrderSessionId(placeOrderResult.Order, orderSessionID);
    ...    Signifyd.Call(placeOrderResult.Order);
    ...    /* Signifyd Modification End */
    ...
    ...    showConfirmation(placeOrderResult.Order);
    ...}
}

```

5.3.2 Modifications to Core Template – Device Fingerprinting

To insert the fingerprint JavaScript snippet in the HTML <head> element, modify the template **default/components/header/htmlhead.isml**.

Add the following lines at the end of the file:

```

<isif condition="${dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydEnableCartridge')}">
    <include template="signifyd_device_fingerprint" />
</isif>

```

The result should look like the following:

```

74
75 <iscomment>Gather device-aware scripts</iscomment>
76 <isinclude url="${URLUtils.url('Home-SetLayout')}"/>
77
78 <!-- Signifyd Modification Start -->
79 <isif condition="${dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydEnableCartridge')}">
80 | <isinclude template="signifyd_device_fingerprint" />
81 </isif>
82 <!-- Signifyd Modification End -->
83

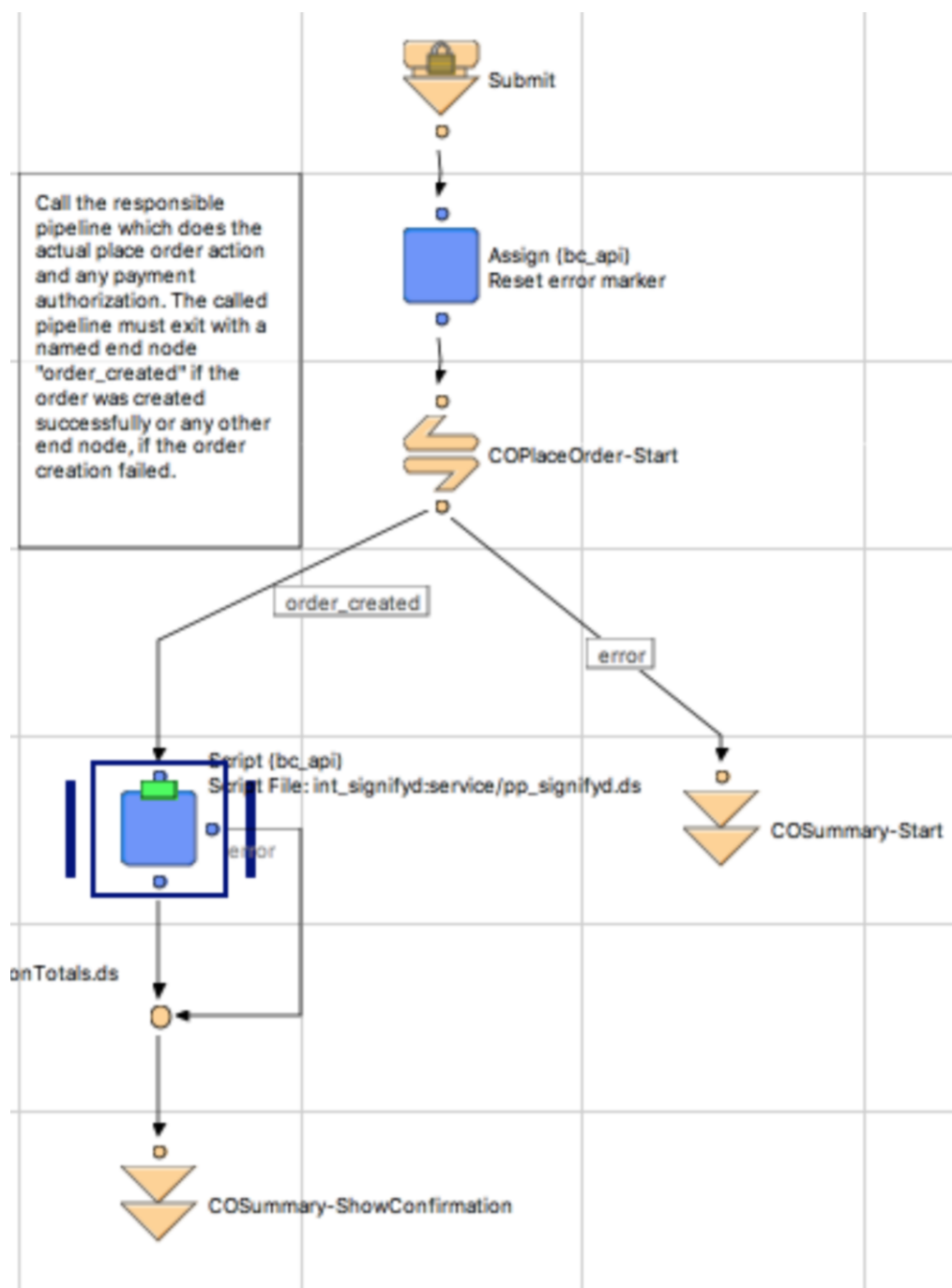
```

5.4 SiteGenesis Integration – Pipelines

Another way to make SFCC send requests to Signifyd is the modification of the default system pipeline for pipeline-based site implementations. Since the pipeline-based approach could eventually be deprecated by Salesforce Commerce Cloud, the controller-based approach is recommended instead.

The Pipeline name is **COSummary**.

Add the script **pp_signifyd.ds** to the end of the **Submit** pipeline as shown below:



The input parameter for this script must be a current Order.

Properties Problems Console	
Pipelet Node - Script (bc_api)	
Property	Value
▼ Configuration	
OnError	PIPELET_ERROR
ScriptFile	int_signifyd:service/pp_signifyd.ds
Timeout	
Transactional	false
▼ Dictionary Input	
Order	Order
▼ Dictionary Output	
ScriptLog	output
▼ Properties	
Custom Label	
Description	

5.5 Limitations and Constraints

5.5.1 AVS and CVV Response Codes

The Signifyd fraud service relies on transaction data passed back from the payment gateway. If a custom payment gateway is implemented, make sure to pass required information to Signifyd by modifying the *signifyd.js* file as shown below:

The values for AVS and CVV Response Code fields MUST map to standard response codes. See [this document](#) for valid response codes. AVS and CVV values should be updated by the merchant for `getSendTransactionParams()` (in case of Pre-auth enabled) and `getParams()` (in case Post-auth enabled) functions on *signifyd.js* file.

`getSendTransactionParams()`

```
function getSendTransactionParams(order) {
  ....var SignifydCreateCasePolicy = dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydCreateCasePolicy').value;
  ....var cal = new Calendar(order.creationDate);
  ....var paymentInstruments = order.allProductLineItems[0].lineItemCtnr.getPaymentInstruments();
  ....var paymentTransaction = paymentInstruments[0].getPaymentTransaction();
  ....var paymentInstrument = paymentTransaction.getPaymentInstrument();
  ....var paymentProcessor = paymentTransaction.getPaymentProcessor();
  ....var mainPaymentInst = getMainPaymentInst(order.getPaymentInstruments());
  ....var mainTransaction = mainPaymentInst.getPaymentTransaction();
  ....var mainPaymentProcessor = mainTransaction.getPaymentProcessor();
  ....var paramsObj = {
  ....  transactions: [{
  ....    transactionId: paymentTransaction.transactionID,
  ....    gatewayStatusCode: '', // to be updated by the merchant
  ....    paymentMethod: paymentInstrument.getPaymentMethod(),
  ....    amount: paymentTransaction.amount.value,
  ....    currency: paymentTransaction.amount.currencyCode,
  ....    createdAt: StringUtils.formatCalendar(cal, "yyyy-MM-dd'T'HH:mm:ssZ"),
  ....    verifications: {
  ....      avsResponseCode: '', // to be updated by the merchant
  ....      cvvResponseCode: '', // to be updated by the merchant
  ....    }
  ....  }],
  ....};
}
```

getParams()

```
        cardBrand: mainPaymentInst.CreditCardType
    },
    amount: mainTransaction.amount.value,
    currency: mainTransaction.amount.currencyCode,
    gateway: mainPaymentProcessor ? mainPaymentProcessor.ID : null
  }];

  if (SignifydCreateCasePolicy === "POST_AUTH") {
    paramsObj.transactions[0].transactionId = mainTransaction.transactionId;
    paramsObj.transactions[0].gatewayStatusCode = ""; // to be updated by the merchant
    paramsObj.transactions[0].paymentMethod = mainPaymentProcessor.ID;
    paramsObj.transactions[0].verifications = {
      avsResponseCode: '', // to be updated by the merchant
      cvvResponseCode: '', // to be updated by the merchant
    };
  }
}

return paramsObj;
```

5.5.2 Order Channel

Order channel information should be updated by the merchant in the **getParams()** function:

```
function getParams(order) {
  .... var SignifydCreateCasePolicy = dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydCreateCasePolicy').value;
  .... var SignifydDecisionRequest = dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydDecisionRequest').value;
  .... var SignifydPassiveMode = dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydPassiveMode');
  .... var orderCreationCal = new Calendar(order.creationDate);
  .... var paramsObj = {
    .... device: {
    ....   .... clientIpAddress: order.remoteHost,
    ....   .... sessionId: order.custom.SignifydOrderSessionId
    .... },
    .... merchantPlatform: getMerchantPlatform(),
    .... signifydClient: getSignifydClient(),
    .... transactions: [],
    .... orderId: order.currentOrderNo,
    .... purchase: {
    ....   .... createdAt: StringUtils.formatCalendar(orderCreationCal, "yyyy-MM-dd'T'HH:mm:ssZ"),
    ....   .... orderChannel: "", // to be updated by the merchant
    ....   .... totalPrice: order.getTotalGrossPrice().value,
    ....   .... currency: dw.system.Site.getCurrent().getDefaultCurrency(),
    ....   .... confirmationEmail: order.getCustomerEmail(),
    ....   .... products: getProducts(order.productLineItems),
    ....   .... shipments: getShipments(order.shipments),
    ....   .... confirmationPhone: order.getDefaultShipment().shippingAddress.phone,
    ....   .... totalShippingCost: order.getShippingTotalGrossPrice().value,
    ....   .... discountCodes: getDiscountCodes(order.getCouponLineItems()),
    ....   .... receivedBy: order.createdBy !== 'Customer' ? order.createdBy : null
    .... },
    .... userAccount: getUser(order),
    .... coverageRequests: SignifydDecisionRequest === "GUARANTEE" ? ["FRAUD"] : (SignifydDecisionRequest === "DECISION" ? ["NONE"] : null)
  .... };
}
```

5.5.3 Fulfillment API

For the Fulfillment API, the following request fields should be updated by the merchant on the function **getSendFulfillmentParams()**, according to the merchant's shipping carrier and following the format specified on the [API documentation](#):

```
function getSendFulfillmentParams(order) {
    var fulfillmentStatus = order.getShippingStatus().displayValue === "PARTSHIPPED" ? "PARTIAL" : "COMPLETE";
    var shipments = order.getShipments();

    var paramsObj = {
        orderId: order.orderNo,
        fulfillmentStatus: fulfillmentStatus,
        fulfillments: []
    };

    var iterator = shipments.iterator();
    while (iterator.hasNext()) {
        var shipment = iterator.next();
        paramsObj.fulfillments.push({
            shipmentId: shipment.shipmentNo,
            // fulfillmentMethod: '', // to be updated by the merchant
            // shippedAt: '', // to be updated by the merchant
            // shipmentStatus: '', // to be updated by the merchant
            // trackingUrls: '', // to be updated by the merchant
            // trackingNumbers: '', // to be updated by the merchant
            // carrier: '', // to be updated by the merchant
            products: getProductLineItems(shipment.productLineItems),
            destination: {
                fullName: shipment.shippingAddress.fullName,
                organization: shipment.shippingAddress.companyName,
                address: getDeliveryAddress(shipment),
                confirmationPhone: shipment.shippingAddress.phone
            }
        });
    }

    return paramsObj;
}
```

5.5.4 Pre-auth default error message

When an order is failed during Pre-auth flow, the default SFCC technical error message is displayed. The merchant can customize the code to display a different error message on the code below:

```

122 .....if (SignifydCreateCasePolicy === "PRE_AUTH")-{
123 .....var SignifydPassiveMode = dw.system.Site.getCurrent().getCustomPreferenceValue('SignifydPassiveMode');
124 .....Signifyd.setOrderSessionId(order, orderSessionID);
125 .....var response = Signifyd.Call(order);
126 .....
127 .....if (!empty(response) && response.declined) {-{
128 .....Transaction.wrap(function() {-{
129 .....if (response.declined) {-{
130 .....order.custom.SignifydOrderFailedReason = Resource.msg('error.signifyd.order.failed.reason', 'signifyd', null);
131 .....}
132 .....if (!SignifydPassiveMode) {-{
133 .....OrderMgr.failOrder(order);
134 .....}
135 .....});
136 .....if (!SignifydPassiveMode) {-{
137 .....res.json({
138 .....error: true,
139 .....errorMessage: Resource.msg('error.technical', 'checkout', null)
140 .....});
141 .....return next();
142 .....}
143 .....}
144 .....}

```

5.5.6 Digital items

If the merchant is selling digital items, the **itemIsDigital** field inside the purchase.products object should be updated to true. By default, the value is set to false.

```

function getProducts(products) {
    var result = [];
    for (var i = 0; i < products.length; i++) {
        var product = products[i];
        var primaryCat = product.product.getPrimaryCategory();

        // get master product's primary category if variant doesn't have one
        if (empty(primaryCat) && !product.product.isMaster()) {
            primaryCat = product.product.masterProduct.getPrimaryCategory();
        }
        var parentCat = !empty(primaryCat) ? primaryCat.getParent() : null;

        result.push({
            itemId: product.productID,
            itemName: product.productName,
            itemUrl: URLUtils.abs('Product-Show', 'pid', product.productID).toString(),
            itemQuantity: product.quantityValue,
            itemPrice: product.grossPrice.value,
            itemSubCategory: !empty(primaryCat) ? primaryCat.ID : null,
            itemCategory: !empty(parentCat) ? parentCat.ID : (!empty(primaryCat) ? primaryCat.ID : null),
            itemImage: product.product.getImage('large', 0).getAbsURL().toString(),
            shipmentId: product.shipment.shipmentNo,
            itemIsDigital: false // to be updated by the merchant in case of digital item
        });
    }
    return result;
}

```


5.5.7 Vanity hostnames

Salesforce will block any access to the hyphenated demandware.net (e.g. development-xxx.demandware.net) that does not originate from the platform itself, which means that any attempt from third-party integrations that access the Storefront through a link will not be able unless passed through a vanity hostname, such as yourbrand.com, www.yourbrand.com, etc. This should be taken into consideration while configuring the URLs for the webhooks in the Signifyd console.

For the Development and Production instances, it is mandatory to use the vanity hostname, so you will need to add an URL Rule like below, for the Staging and sandbox instances this way can be used as well, but it is not mandatory.

SEO & Discoverability > URL Rules > Pipeline URLs

General Settings Catalog URLs Content URLs **Pipeline URLs** URL Search Refinements

Pipeline URLs

Use the table below to define one or multiple aliases for any pipeline used in your Storefront.

Pipeline URL Settings

Append Trailing Slash to Pipeline URLs: ☐ Append Trailing Slash to Pipeline URLs

Perform Automatic Redirect: ☐ Enable this to automatically redirect requests to the desired trailing slash configuration.

Pipeline Aliases

Alias or Pipeline: Locale:

[Select All](#)

Alias

Pipeline

☐

signifydcallback

resolves To

Signifyd-Callback

New Alias:

resolves To

And add e.g. <https://yourStoreURL.com/s/SiteGenesis/signifydcallback> to the Webhook Addresses in <https://app.signifyd.com/settings/notifications>

If the custom site preference **SignifydCreateCasePolicy** is set to 'PRE_AUTH', there is no need to configure a webhook to receive the decision back, unless the merchant needs aforementioned Signifyd's manual review services.

5.6 Updating from V2 to V3

Merchants who are upgrading from the V2 API to the V3 API should install the V3 compatible cartridge on top of their existing V2 cartridge. This installation process will overwrite the existing cartridge files. It is important to import all files in the metadata folder to ensure that the respective metadata, jobs, and services are updated to be compatible with the V3 API.

Please keep in mind that any customizations made by the merchant will need to be reapplied on top of the V3 cartridge files after the installation is complete.

Any outdated services and custom preferences from the V2 cartridge that will not be used in the V3 version can be safely removed once the successful installation of the V3 cartridge has been confirmed.

The outdated V2 services that can be removed are shown on the image below:

☐	Signifyd.REST.CreateCase
☐	Signifyd.REST.SendFulfillment
☐	Signifyd.REST.SendTransaction

In V3, those services were replaced with new services that reflect the V3 API structure:

- SignifydCheckout
- SignifydSale
- SignifydSendFullfilment
- SignifydTransaction

5.7 Logging

You can enable logs for all operations with the Signifyd API for debugging. But don't forget to disable it after debugging to prevent uncontrolled growth of log files.

Go to [Administration > Operation > Custom Log Settings](#). You can enable specific levels of logging for Signifyd. Each level brings a different or higher level of detail in the logs

[Administration](#) / [Operations](#) /
Custom Log Settings ⓘ

On this page you can configure the writing of custom code log messages. System log messages aren't configurable. The changes you make will become active only when you click "Save".

Custom Log Filters

You may only configure log categories for custom and service logging. While the categories of those messages start with 'custom.' in Log Center, you must not use this prefix in category names.

Active	Category Prefix	Log Category	Log Level	
	custom.	Enter a log category...	WARN ▾	Add
		root	DEBUG ▾	
☒	custom.	signifyd	DEBUG ▾	

Note: DEBUG log level is only available in sandboxes, Development and Staging instances. For Production instances, any log level above DEBUG can be used.

5.8 Configurations on Signifyd Console

All Signifyd cases created during SFCC order creation can be viewed here: <https://app.signifyd.com/cases>.

API key, Profile and all other settings can be set up on this page: <https://app.signifyd.com/settings>.

The ORDER_CHECKPOINT_ACTION_UPDATE webhook should be setup following the instructions on <https://docs.signifyd.com/#operation/CreateTeamWebhook> in order to update SFCC with the latest status from Signifyd. Webhooks can be set up directly in the console by going to <https://app.signifyd.com/settings> >> Integrations >> and configuring the v3 ORDER_CHECKPOINT_ACTION_UPDATE webhook for the given environments.

5.9 Testing

5.9.1 Pre-auth flow

Pre-auth Test Case 1: Signifyd response: Accept

Expected Result:

1. Before the payment processor authorization, a synchronous call to Checkout API is made and a case is created on Signifyd.
2. Signifyd sends the checkpointAction field as “Accept” in the response. The attributes **SignifydCaseID**, **SignifydOrderURL**, **SignifydFraudScore**, **SignifydPolicy** and **SignifydPolicyName** are saved on the order and the order proceeds to authorization.
3. Depending on the payment processor response:
 - a. Authorized: order proceeds to be placed. A Transaction API call is made after the order is placed to update the order details in Signifyd.
 - b. Not authorized or error: the order fails. A Transaction API call is made to update the order details in Signifyd.

Pre-auth Test Case 2: Signifyd response: Reject and Passive Mode: No.

Expected Result:

1. Before the payment processor authorization, a synchronous call to Checkout API is made and a case is created on Signifyd.
2. Signifyd sends the checkpointAction field as “Reject” in the response. The attributes **SignifydCaseID**, **SignifydOrderURL**, **SignifydFraudScore**, **SignifydPolicy** and **SignifydPolicyName** are saved on the order.
3. The order fails and the attribute **SignifydOrderFailedReason** is updated on the order.
4. The default error message is shown to the user on the checkout page

Pre-auth Test Case 3: Signifyd response: Reject and Passive Mode: Yes.

Expected Result:

1. Before the payment processor authorization, a synchronous call to Checkout API is made and a case is created on Signifyd.
2. Signifyd sends the checkpointAction field as “Reject” in the response. The attributes **SignifydCaseID**, **SignifydOrderURL**, **SignifydFraudScore**, **SignifydPolicy** and **SignifydPolicyName** are saved on the order.
3. The attribute **SignifydOrderFailedReason** is updated on the order and the order proceeds to be authorized.
4. Depending on the payment processor response:

- a. Authorized: order proceeds to be placed. A Transaction API call is made after the order is placed to update the order details in Signifyd.
- b. Not authorized or error: the order fails. A Transaction API call is made to update the order details in Signifyd.

Pre-auth Test Case 4: Signifyd response: Hold

Expected Result:

1. Before the payment processor authorization, a synchronous call to Checkout API is made and a case is created on Signifyd.
2. Signifyd sends the checkpointAction field as “Hold” in the response. The attributes **SignifydCaseID**, **SignifydOrderURL**, **SignifydFraudScore**, **SignifydPolicy** and **SignifydPolicyName** are saved on the order.
3. Depending on the payment processor response:
 - a. Authorized: order proceeds to be placed. A Transaction API call is made after the order is placed to update the order details in Signifyd.
 - b. Not authorized or error: the order fails. A Transaction API call is made to update the order details in Signifyd.
4. ORDER_CHECKPOINT_ACTION_UPDATE Webhook will be triggered to send a new checkpointAction and update the order **SignifydPolicy** and **SignifydPolicyName** attributes.

5.9.2 Post-auth flow

Post-auth Test Case 1: Signifyd response: Accept

1. After the order is successfully placed, a call to Sale API is made.
2. The attributes **SignifydCaseID** and **SignifydOrderURL** are saved on the order.
3. The ORDER_CHECKPOINT_ACTION_UPDATE Webhook is triggered, a callback is made to SFCC to update the order attribute **SignifydPolicy** with the “Accept” checkpointAction value from the response.
4. If **SignifydHoldOrderEnable** custom preference is set to Yes, the order export status is updated to Ready to Export.

Post-auth Test Case 2: Signifyd response: Reject

1. After the order is successfully placed, a call to Sale API is made.
2. The attributes **SignifydCaseID** and **SignifydOrderURL** are saved on the order.
3. The ORDER_CHECKPOINT_ACTION_UPDATE Webhook is triggered, a callback is made to SFCC to update the order attribute **SignifydPolicy** with the with the “Reject” checkpointAction value from the response.
4. If **SignifydHoldOrderEnable** custom preference is set to Yes, the order export status is updated to Not Exported.

5.9.3 Automated Testing

On this implementation we have provided unit and integration testing.

5.9.3.1 Unit testing

To do the automated testing open your command prompt (Windows) or terminal (MacOS/Linux/Unix) navigate to the folder above the cartridges and run the command “npm run test”, this will test the functionalities within the cartridge source code.

5.9.3.2 Integration testing

To run the automated integration testing create a dw.json file and navigate to the folder above the cartridges

```
{  
  "hostname" : "somesb.demandware.net",  
  "username" : "someUser",  
  "password" : "somePassword",  
  "version" : "someversion"  
}
```

Then run the command npm run test:integration

5.10 Troubleshoot

5.10.1 Missing API Key

If the API key provided by Signifyd is not correctly informed in the site custom preferences, the following error will be seen in the logs for any of the services:

Null Arguments in The given string to encode was null

Solution: The merchant should verify if the custom preference **SignifydApiKey** is correctly setup with the API key.

5.10.2 Wrong API Key

If the API key is wrong, the following error will be seen in the logs for any of the services:

..Provided API Key not found.

Solution: The merchant should double check if the custom preference **SignifydApiKey** has the correct API key.

5.10.3 Service not enabled

If a service is not correctly enabled in Business Manager, the following error will be triggered:

Service={name_of_the_service} status=SERVICE_UNAVAILABLE errorCode=0 errorMessage=The service is not enabled

Solution: The merchant should double check if the service informed in the error message is enabled.

5.10.4 Order fields not being updated

If the Signifyd related fields in the orders are not being updated, the possible reasons are:

- **Incorrect setup of the code.** The merchant should check if all steps to integrate the cartridge were correctly followed.
- **Incorrect setup of webhooks.** The merchant should check if only the ORDER_CHECKPOINT_ACTION_UPDATE webhook is being used and correctly setup in the Signifyd console.
- **Code customizations.** The merchant should check if any customizations applied are not interfering in the cartridge functionality.
- **Incorrect data being passed in the service requests.** Some of the fields for Transaction and Fulfillment API need to be manually mapped by the merchant directly in the code during the integration. If a field is not correctly mapped or has a wrong value, this might cause the API to not work correctly, thus not updating the order.
- **Service timeouts.** The service timeout can be increased in the service configurations if needed, following SFCC default steps to do it. If the timeouts persist, Signifyd should be contacted.

5.10.5 Webhook Callbacks Are Blocked by SFCC

If a merchant has the **Limit Storefront Order Access** preference under **Merchant Tools > Site Preferences > Order > Order Access Settings** set as **Allow List**, Signifyd webhook callbacks might be blocked, if the Signifyd-Callback endpoint is not added to the allow list. Verify the Limit Storefront Order Access allow list in your SFCC instance to make sure that the Signifyd-Callback endpoint is added to the list of allowed endpoints. This list controls which endpoints can access and modify order data.

6 Operations, Maintenance

6.1 Availability

Availability/Uptime is 24/7 is the intended access. But in case of service failure the order will get placed. And a script step job can be configured to create the missing orders not sent to Signifyd.

6.2 Support

For implementation questions or issues please contact your assigned Customer Success Manager. For general support questions or issues [contact Support](#).

6.3 Intended Locales

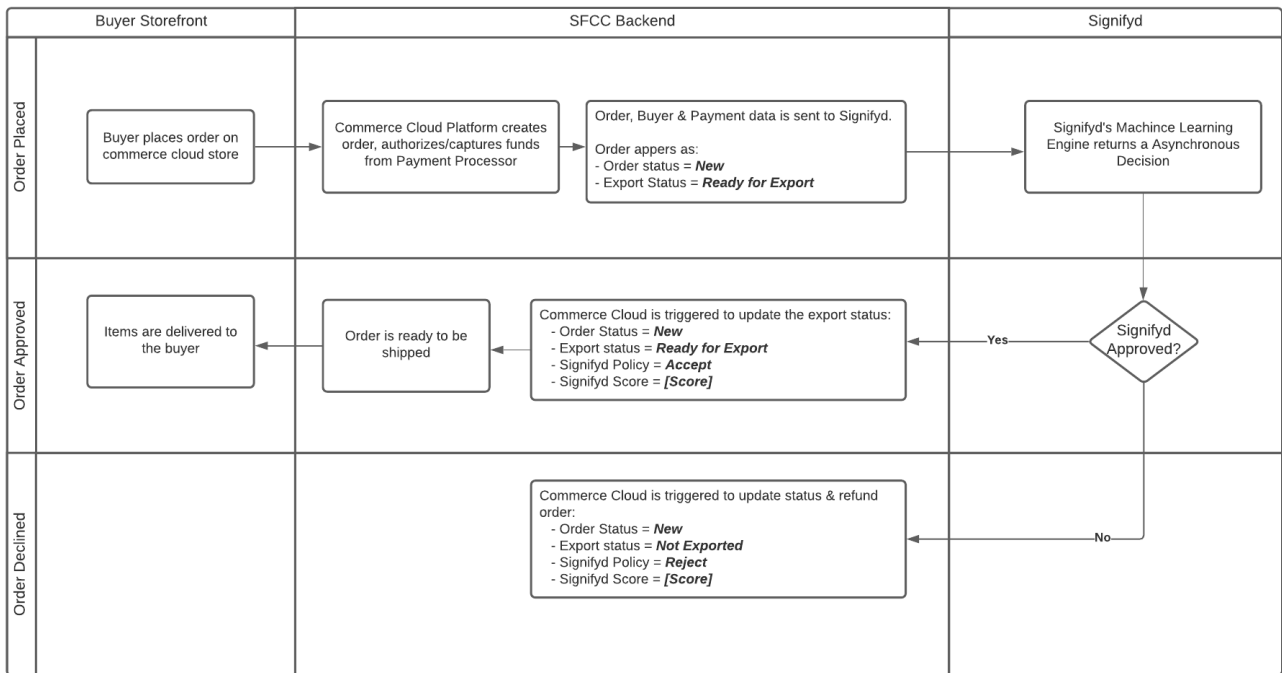
Out of the box the cartridge supports “en_US” locale, but other locales may be added according to Signifyd’s service availability, Signifyd has support for multiple countries and locales.

7 Release History

Version	Date	Changes
16.1.0	08/15/2015	Initial release
18.1.0	04/01/2018	Added Job Schedule configuration and template modification
19.1.0	02/28/2019	Added modifications to support SFRA
20.1.0	03/24/2020	Added Unit and Integration testing
22.1.0	05/31/2022	V3 API release

8 Process Flow Diagrams

SFCC Post-Auth Flow



SFCC Pre-Auth Flow

